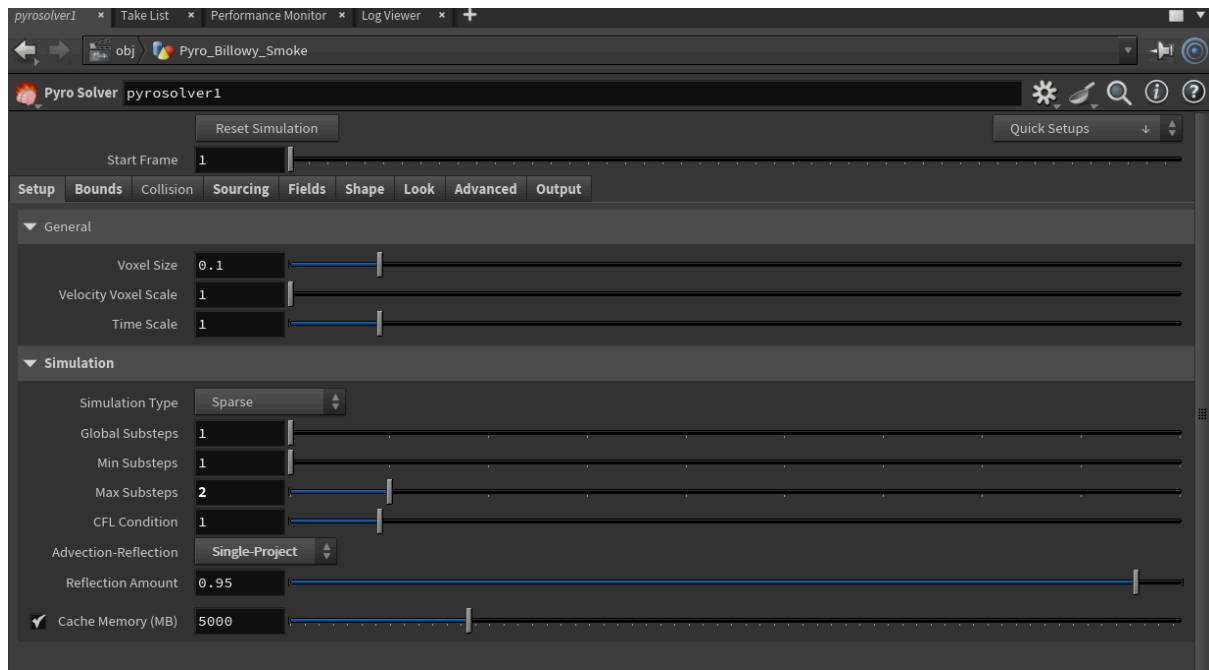


Simple Guide on using Venkys Turbulence Microsolver

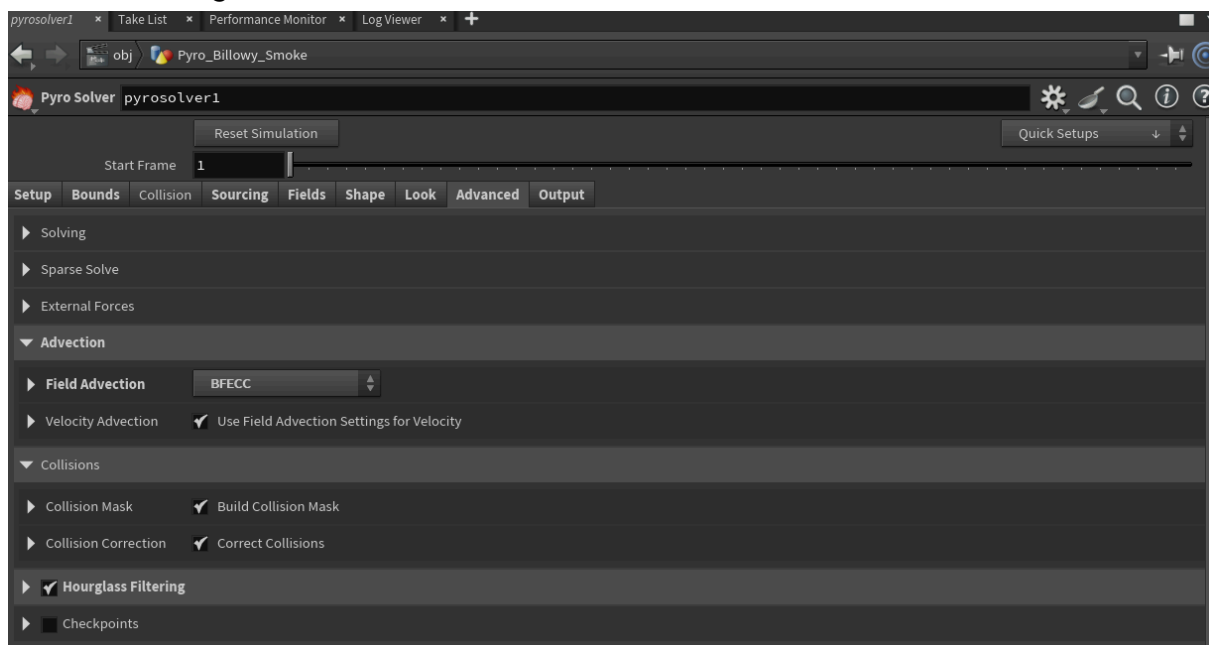
Here I am using the Pyro Billowy Smoke shelf tool in H21.

When I first load the preset, I make a couple of main adjustments that help with the overall precision of the pyro solver.

First, I use 2 max substeps with *Single Project* in Advection-Reflection.

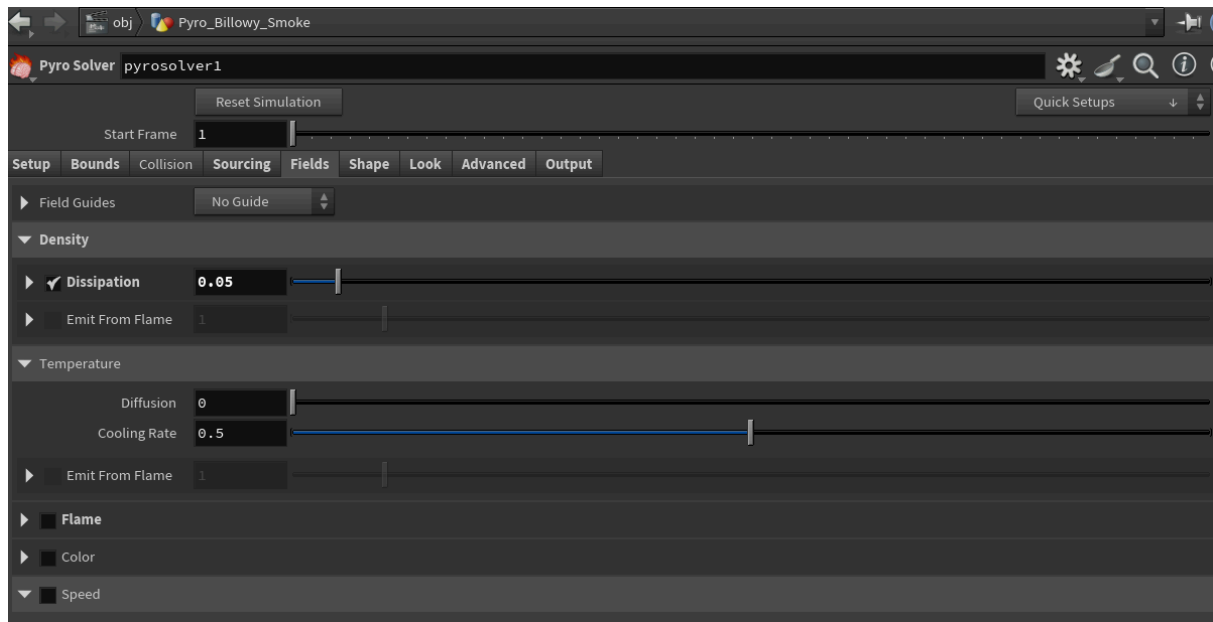


Second, I change the Advection Scheme to **BFEC**.

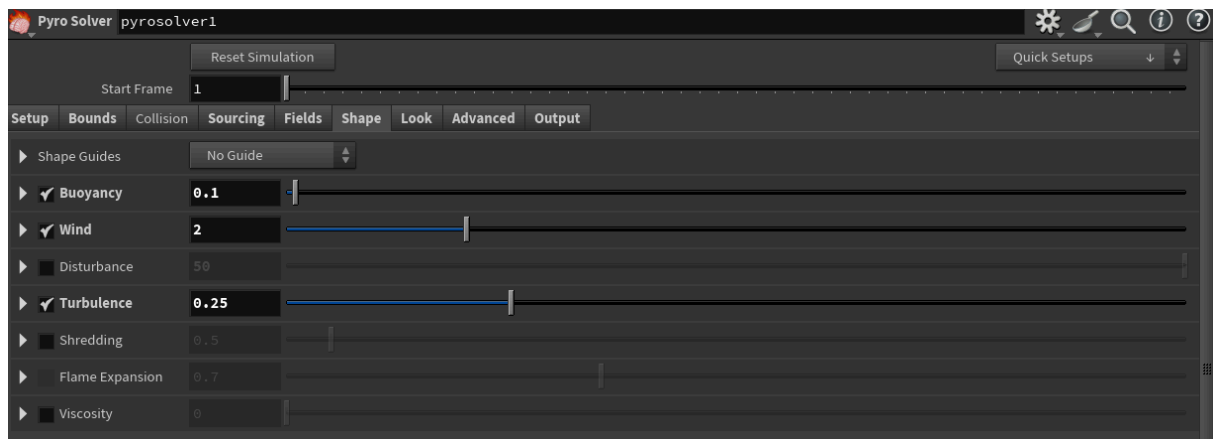


I also uncheck Hourglass Filtering and disable all other output fields except density for faster viewport performance.

To optimize the solver, I uncheck *Flame* and *Speed* in the Fields section (since we are not simulating fire).

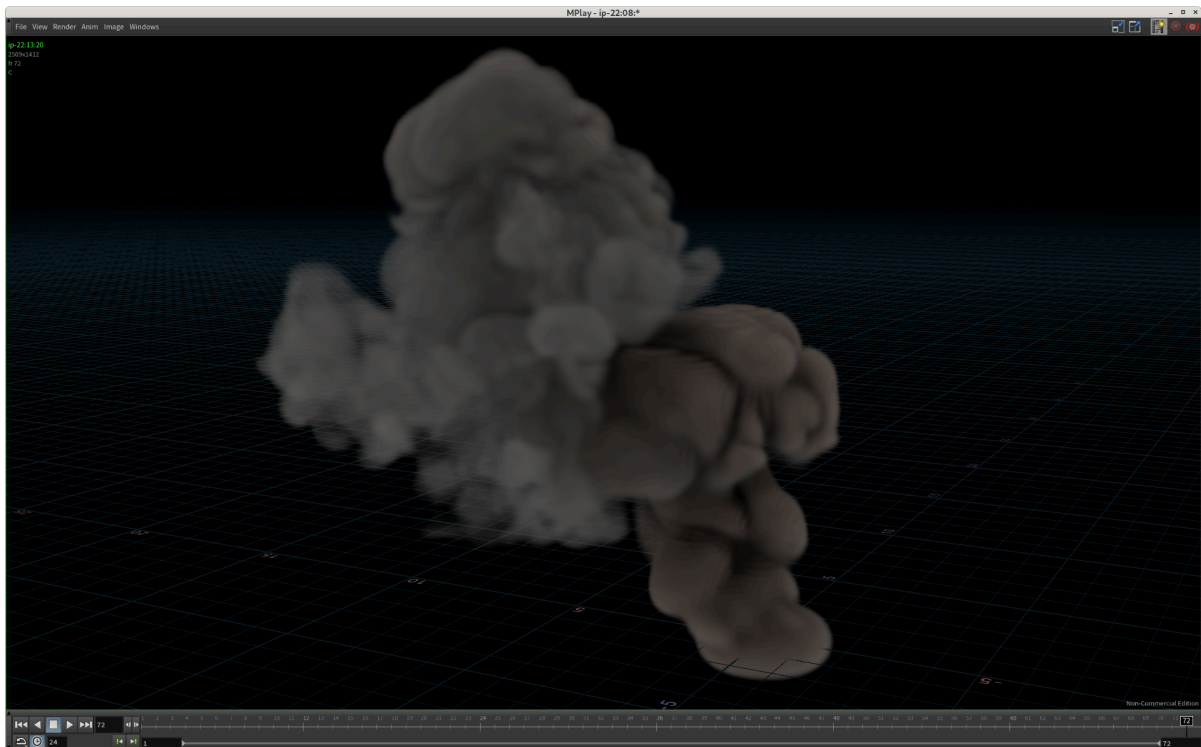


Since our solver is replacing the Disturbance, I also uncheck it in the Shape section.

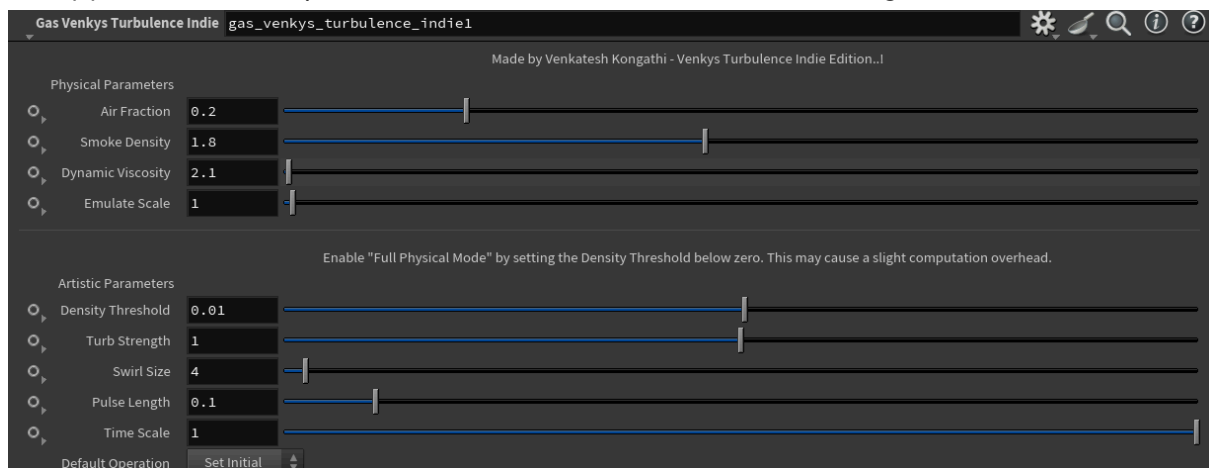


I made a flipbook of 72 frames with the current settings for reference, let's call it a vanilla flipbook.

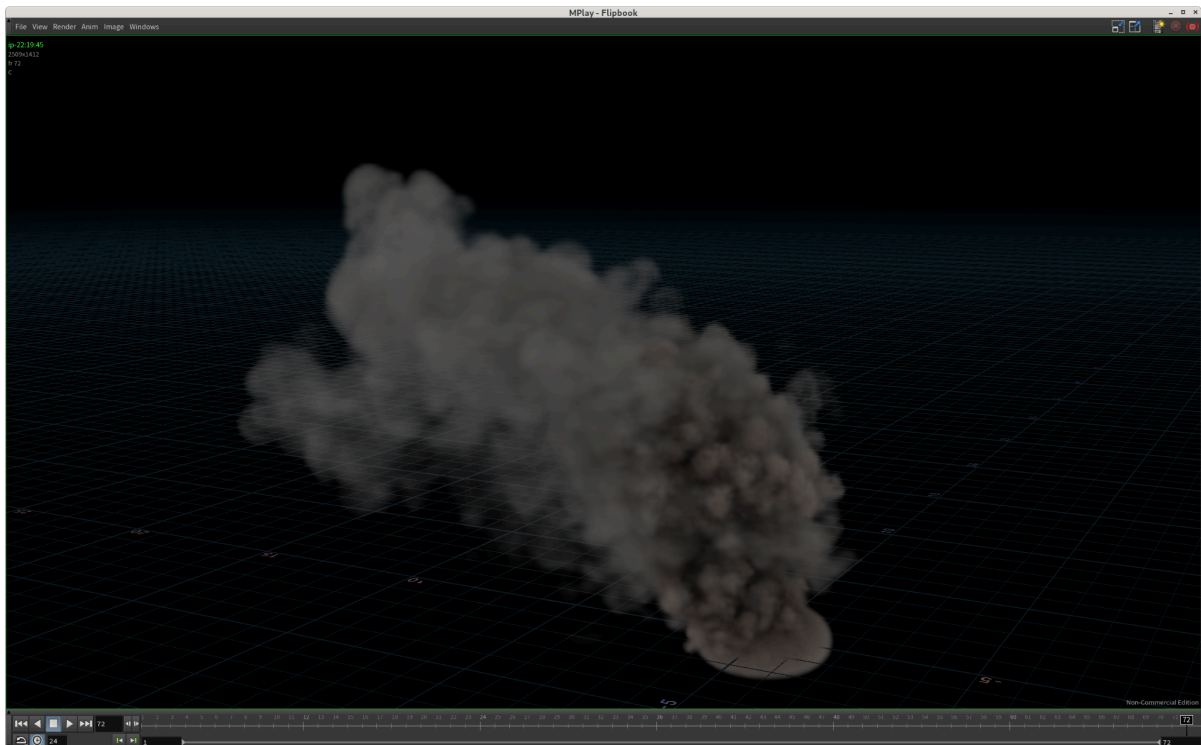
It gives a good idea of which physical parameters you need to tweak in Venky's Turbulence microsolver.



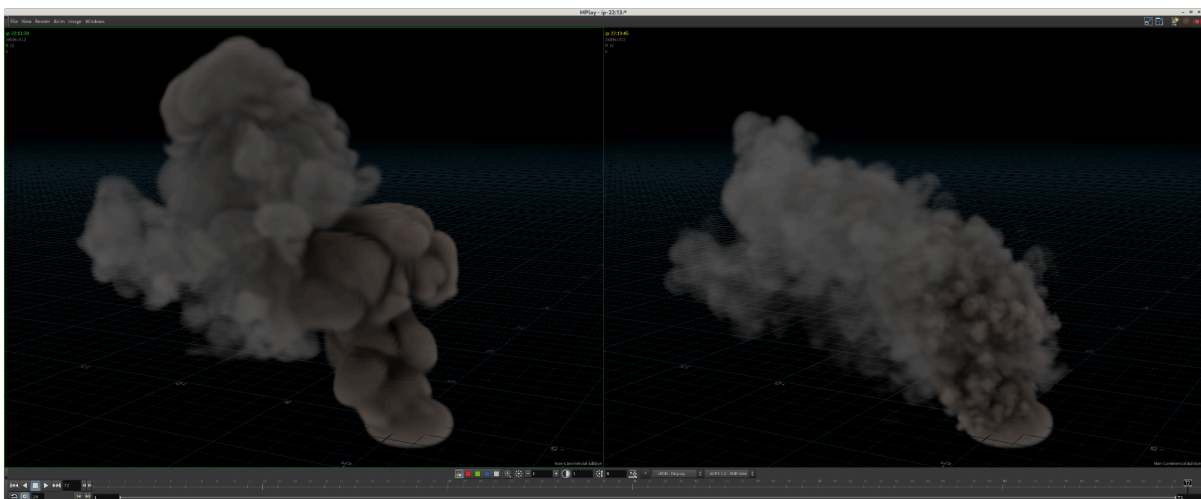
I dropped down Venky's Turbulence solver with its default settings.



I quickly ran a new flipbook sequence to see its effect. It doesn't look that nice!



If we compare this with the vanilla flipbook, it doesn't match the silhouette. This means our dynamic viscosity value is not matching the scale of the smoke we are trying to simulate.

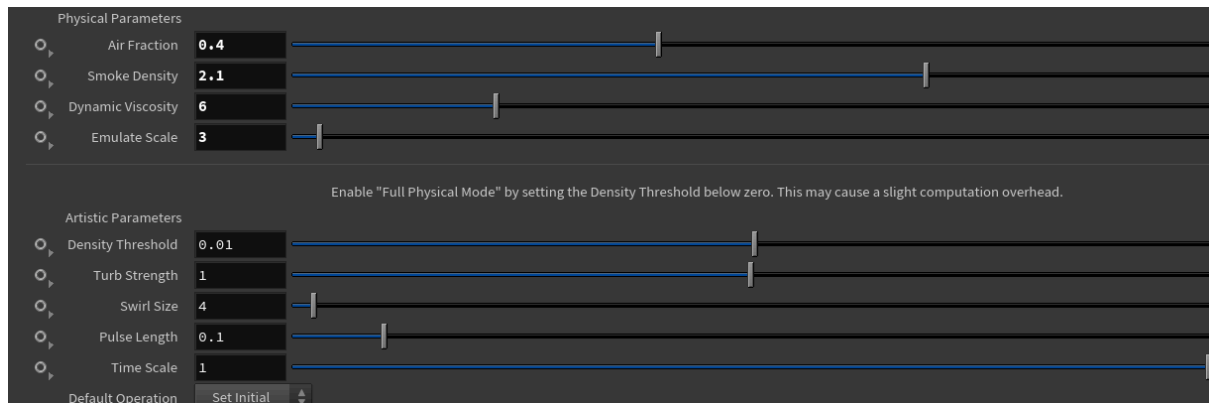


Since we are using a density threshold above zero, the turbulence is partially physical, so I raised the Air Factor to 0.4 for better breakup.

Our smoke looks quite dense, so I increased the smoke density to 2.1 to match the look.

Based on the vanilla flipbook, the smoke travels almost 10 meters across and 6 meters high.

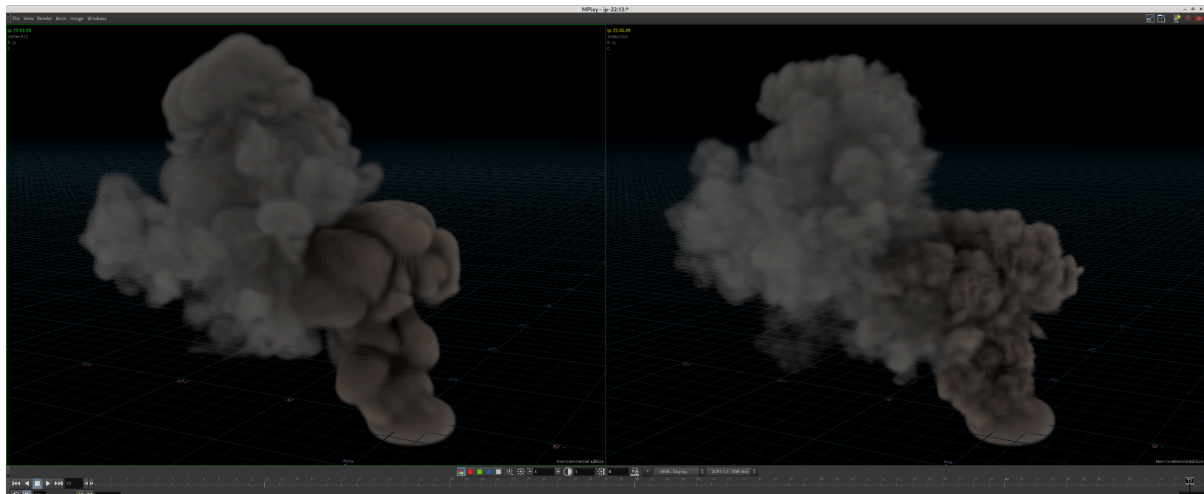
This means our dynamic viscosity needs to be higher, so we can use a value around 6 and emulate scale around 3.



I then ran a new flipbook sequence to compare with vanilla.

As you can see, the smoke almost matches the vanilla version in terms of silhouette while adding a ton of organic detail.

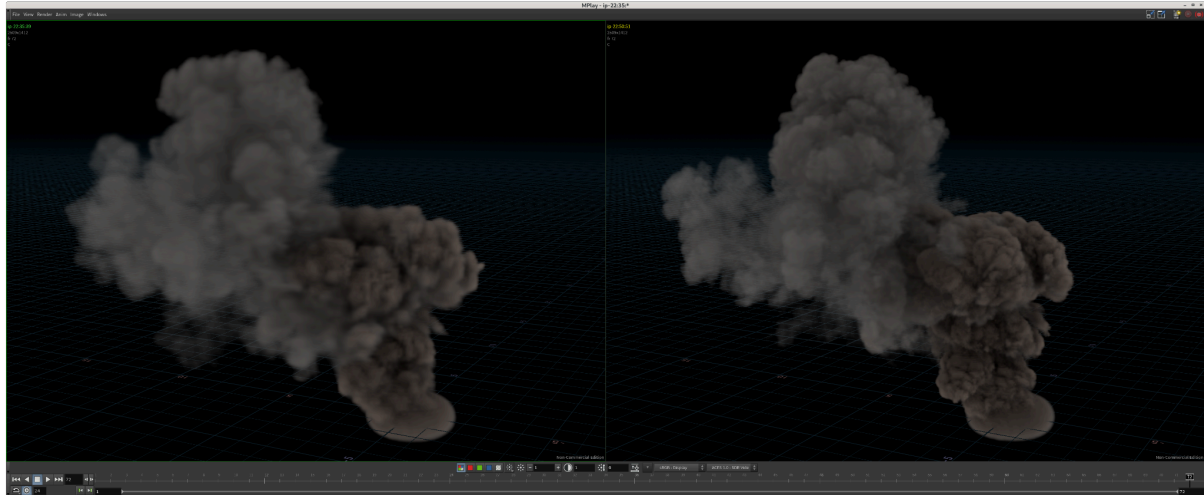
Since our turbulence adds natural air drag, it won't exactly grow or match the vanilla shape.



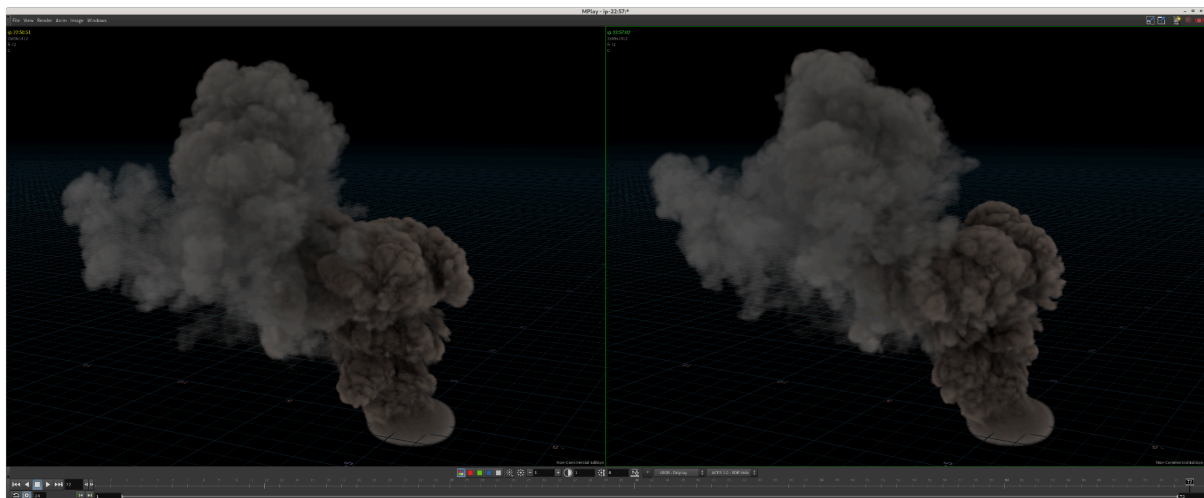
As you can see, it's quite simple to use once you have the vanilla sim for reference.

I changed the voxel resolution from 0.1 to 0.05 and ran a new flipbook.

As you can see, it produces the maximum amount of detail for that resolution while maintaining the silhouette.



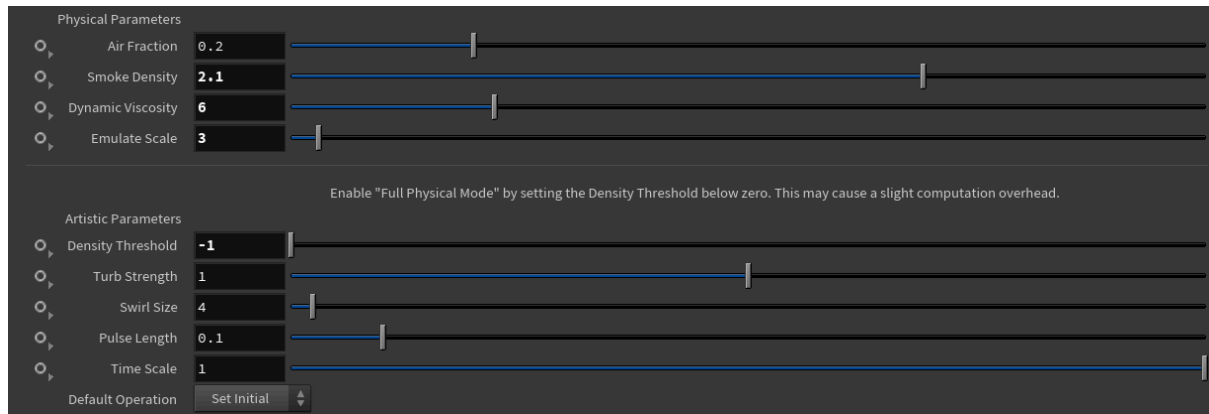
I also did a flipbook with 3 max substeps to show the difference in detail. Pick whichever you prefer.



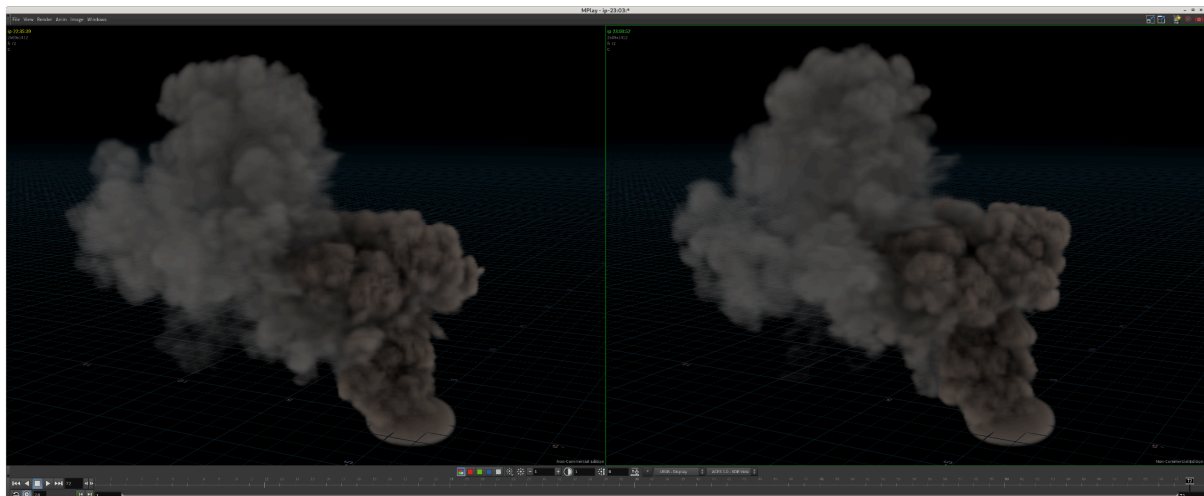
Now I will go back to the lower resolution sim with a voxel size of 0.1 and change max substeps to 2.

Let's look into full physical mode.

For that, bring down the Air Factor to 0.2 and set the density threshold to -1.

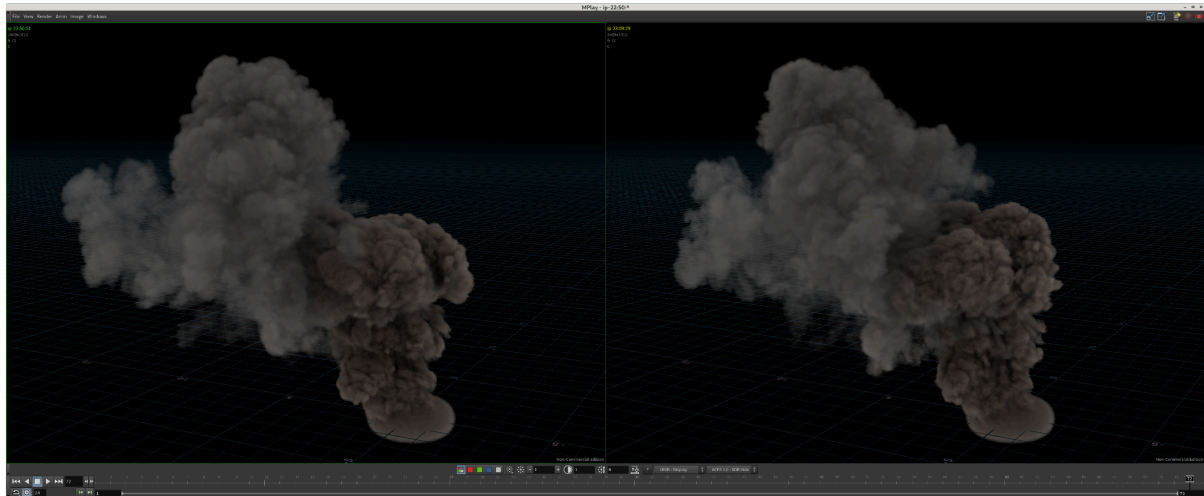


Full physical mode (right) has more surface detail and adds natural diffusion to the smoke as it rises, making it puffier. You can see this in the comparison with partial mode (left).



I also ran a high-resolution version to compare partial physical (left) and full physical (right).

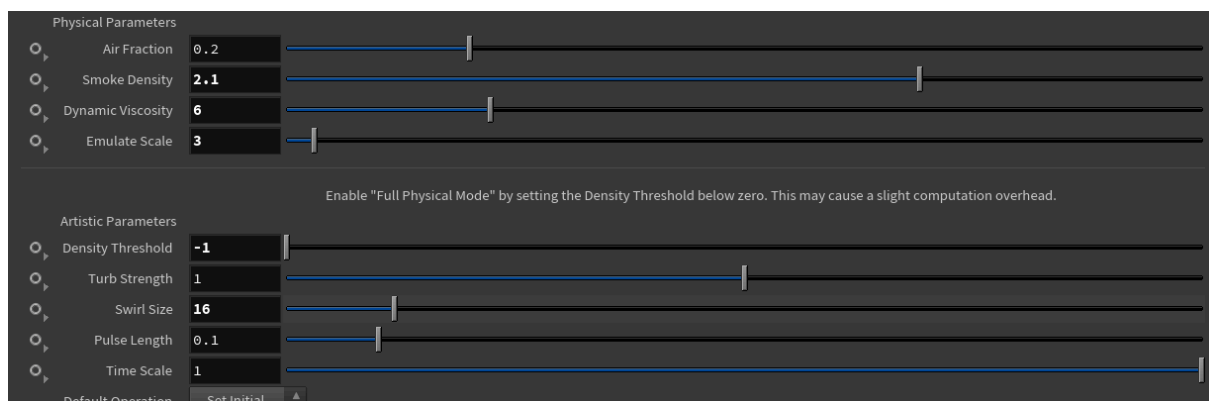
You can notice a lot of surface detail.



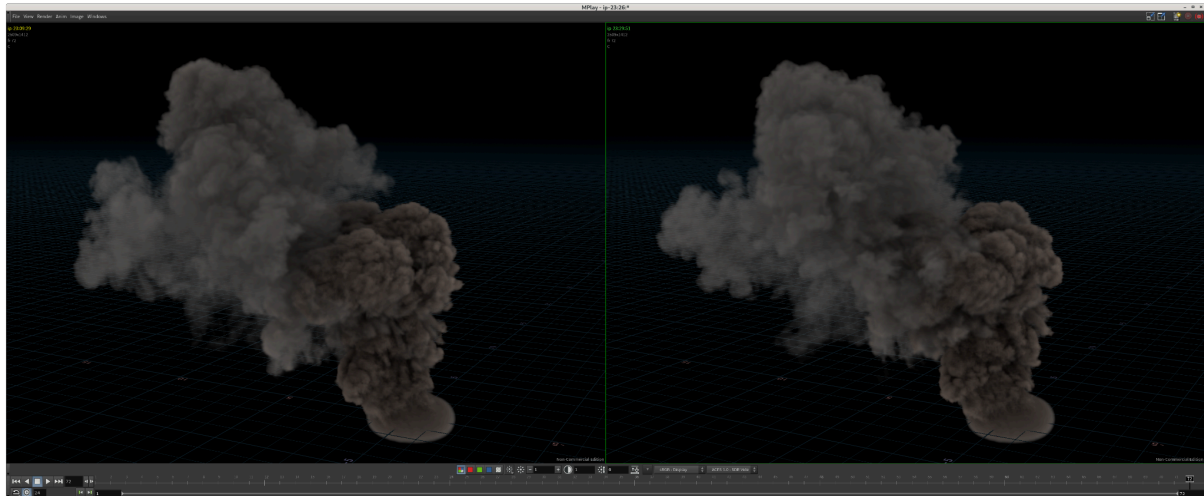
Now let's see what Swirl Size does.

Swirl Size 4 gives maximum detail at a given voxel resolution with minimal artifacts. This helps maintain the silhouette with the vanilla simulation.

Let's change the Swirl Size to 16, which means each swirl occupies around 16 voxels across.



I ran a new flipbook sequence with the same high-res and full physical settings. As you can see, the silhouette and details look a bit different. This way, you can add an interesting breakup to your simulations if needed.



Key Points:

- **Turb Strength:** As you can see, I have never touched the Turb Strength parameter at all. It's there if you want to adjust turbulence slightly, e.g., reduce it by 30% or raise it by 20%.
- If you are doing an explosion or something that occupies a large area, higher values of dynamic viscosity work better, something around 10-20. If your explosion has a lot of expansion and it is not expanding after adding turbulence, it means your dynamic viscosity is too low for the given smoke density value.
- Use *Emulate Scale* to add more detail when you have high dynamic viscosity.
- Use *Pulse Length* to experiment with different looks.
- When the density threshold is positive and you see artifacts, raise it slightly to help remove them.

Thank you for reading through. I hope this helps.

Enjoy!